

Ethics / Consent

The study was conducted in accordance with ethical standards for survey research. Participants were informed about the scientific purpose of the study prior to participation and were notified that withdrawal was possible at any time without consequences. All responses were collected anonymously. No formal ethics committee approval was required, as the study involved a non-sensitive anonymous survey of the general adult population.

Experimental / Manipulation check

These have been addressed in the submission directly.

English survey (translated from German)

The translated English survey can be found under the document name "Survey_Blockchain_in_tourism.pdf"

Sample Quality Checks und A-priori-Power-Reasoning

According to the white paper (https://www.hslu.ch/-/media/campus/common/files/dokumente/h/1-medienmitteilungen-und-news/2026/w/260119-w-ikm-studie-blockchain-whitepaper-24s-web.pdf?sc_lang=de-ch; in German), the sample (n = 1'244) was weighted and is representative in terms of gender, age, language region and place of residence.

The survey was conducted via the DemoSCOPE Community Panel, which, according to the company, comprises approximately 60'000 people and was established through multi-channel recruitment (online, by phone, and in person). DemoSCOPE is also certified to ISO 20252:2019, which attests to an established quality standard for market and social research.

Power Analysis (G-Power), Post-hoc analysis

Post Hoc Power Analysis Procedure

To evaluate the statistical adequacy of the study's sample size (N = 1'244) and to verify the robustness of the hierarchical multiple regression analyses, a post hoc power analysis was conducted. The procedure was carried out using the G*Power software application, version 3.1 ([Faul, Erdfelder, Buchner, & Lang, 2009](#)).

Analytical Setup

The objective of the analysis was to determine the achieved statistical power (1 - beta) of the final interaction models (Model 3) for both experimental use cases. Within the G*Power interface, the test family was set to "F tests" and the specific statistical test selected was 'Linear multiple regression: Fixed model, R² deviation from zero'. The type of power analysis was defined as 'Post hoc: Compute achieved power', given alpha, sample size, and effect size.

Parameter Specification

For both scenarios, the Type I error probability (alpha) was established at the conventional threshold of 0.05 and the total sample size was entered as 1'244. The required effect size metric for this test is Cohen's f². Rather than relying on standard

convention estimations, the exact f^2 values were computed mathematically within G*Power using the observed squared multiple correlations (R^2) from the respective regression models.

Use Case 1

The final hierarchical model contained 11 predictors. Based on the observed R^2 of 0.608, the calculated effect size was $f^2 = 1.551$. The analysis yielded an achieved statistical power of 1.000.

Test family		Statistical test	
F tests		Linear multiple regression: Fixed model, R ² deviation from zero	
Type of power analysis			
Post hoc: Compute achieved power – given α, sample size, and effect size			
Input Parameters		Output Parameters	
Determine =>	Effect size f ²	Noncentrality parameter λ	1929.469
	α err prob	Critical F	1.7964081
	Total sample size	Numerator df	11
	Number of predictors	Denominator df	1232
		Power (1–β err prob)	1.0000000

G*Power output for Use Case 1

Use Case 2

The final hierarchical model contained 9 predictors. Based on the observed R^2 of 0.680, the calculated effect size was $f^2 = 2.125$. The analysis yielded an achieved statistical power of 1.000.

Test family		Statistical test	
F tests		Linear multiple regression: Fixed model, R ² deviation from zero	
Type of power analysis			
Post hoc: Compute achieved power – given α, sample size, and effect size			
Input Parameters		Output Parameters	
Determine =>	Effect size f ²	Noncentrality parameter λ	2643.5
	α err prob	Critical F	1.8874527
	Total sample size	Numerator df	9
	Number of predictors	Denominator df	1234
		Power (1–β err prob)	1.0000000

G*Power output for Use Case 2

Conclusion

For both experimental conditions, the achieved statistical power was > 0.999 . These values exceed the conventional minimum statistical power threshold of 0.80. Consequently, it can be concluded that the sample size of 1'244 was sufficient for the

complexity of the hierarchical regression models and the probability of committing a Type II error in this study is negligible.